Examples and Explorations of Mathematical Explanation in Support of the Indispensability Argument

Within this essay, I aim to contribute to the debate surrounding the so-called indispensability argument by proposing an example of mathematical explanation within science. The indispensability argument is an attempt to demonstrate that we ought to believe in mathematical objects that are not spatio-temporally located and do not interact causally (i.e. abstract mathematical objects) via appeal to the role of mathematics within science. After exploring why examples of mathematical explanation in science support the indispensability argument, I will propose a novel instance of mathematical explanation which invokes reference to Fibonacci numbers within nature. While sceptical of it, I will then entertain the objection that the mathematics directly referenced can be removed from the explanation without loss of explanatory power: the phenomenon in question could be sufficiently explained by referring to evolutionary theory. This will then lead me to consider the theory of evolution, and its nature as a statistical theory. Thus, I will argue that if statistics indispensably refers to abstract mathematical objects, given the widespread use of statistics within scientific theories, and particularly given the use of statistical explanations, this would suggest new evidence in favour of the indispensability argument, and that we ought to believe in abstract mathematical objects. I discuss this in the context of the theory of evolution, but argue on a preliminary basis that further evidence could also be found from more of our best scientific theories, such as within thermodynamics or quantum mechanics. Therefore, I propose that it would be enriching for the indispensability debate for further philosophical work to be done to determine if statistics indispensably uses abstract mathematical objects.

Indispensability and Explanation

To begin, I will briefly outline the indispensability argument using a form commonly found in the philosophical literature (Colyvan, 2001, Chap. 1) (Leng, 2010, Chap. 1) (Maddy, 1997, 133). Reformulations can be found, such as for those with qualms regarding scientific holism (Colyvan, 2001, 13-15), but for the purpose of this essay I will accept and refer to the following conception of it. The first point of the argument is that in order to find out what we ought to believe, we should look to our best scientific theories. This can be referred to as ‘naturalism’. Very roughly, our best scientific theories can be understood as those which are the most

accepted within the scientific community, and explain the most phenomena. Quantum Field Theory and General Relativity are some examples of our best scientific theories within Physics. The second point can be referred to as 'scientific holism': confirmation of a theory as one of our best scientific theories is a confirmation of all statements within that theory, to the same degree.[1] Next, our best scientific theories indispensably require the use of abstract[2] mathematical objects. This term 'indispensable' can mean a few different things, but I think it is most plausible to take it to mean that the abstract mathematical objects within a theory cannot be removed to result in a theory that has the same 'observational consequences' as the first, where the second theory must also be preferable (Colyvan, 2001, 77). This idea of 'preferable' is also open to interpretation, but plausible criteria might be parsimony, unificatory or explanatory power, 'boldness' (i.e. ability to generate novel predictions), and formal elegance (ibid., 78-79). Therefore, finally, as our best scientific theories indispensably use abstract mathematical objects, and tell us what we ought to believe in, we ought to believe that there are abstract mathematical objects. This concludes the indispensability argument.

In order to justify the philosophical importance of mathematical explanation in science, I will first further discuss what is required for indispensability via an example of dispensable mathematics. Referring back to what it means for something to be indispensable to a scientific theory, it does seem that there are some instances where mathematics could be removed without affecting observational consequences. Take the case of complex numbers. These are made up of a so-called 'real' part (what mathematicians refer to as a real number, meaning it can be represented by a number line) and an 'imaginary' part, which is a multiple of *i*, the square root of -1. For example,

$$x = 3 + 4i = 3 + 4\sqrt{-1}$$

demonstrates a complex number *x*. Without a rigorous exposition of the mathematics, certain complex numbers will multiply together to produce real numbers, and they allow for mathematical manipulations of sine and cosine functions via use of exponentials. An argument can therefore be made that talk of complex numbers is used by scientists merely in order to speed up calculations and make faster inferences, not to tell us what we ought to believe. Therefore, they are not indispensable to scientific theory: they can be removed without

[1] Other versions of the indispensability argument may replace this point with appeal to inference of the best explanation (Baker, 2009, 613) or to justifications of scientific practice being justifications of the use of mathematics (Colyvan, 2001, 14-15).

[2] I will not address why the objects are necessarily abstract within this essay, but take it as given.

affecting observational consequences, and removing them means the theory quantifies over fewer abstracta, so is a more parsimonious (and therefore better) theory.[3]

How, then, should we identify when mathematics is inarguably indispensable to scientific theory? This is where the role of mathematical explanations within science becomes important. Work by Mark Colyvan and Joseph Melia suggests that this would be best done by finding an example of where the mathematics within science is truly *explanatory* (Baker, 2005, 224). If this condition is met, it seems difficult to argue that the mathematics is dispensable, if it is required to explain some phenomenon. By explain, I mean perhaps to make a phenomenon less mysterious (Colyvan, 2001, 46-47), to infer a phenomenon from certain premises which include general laws of nature (Baker, 2005, 235), or to answer a why-question (ibid.). Any theory of explanation that does not rely on causality is sufficient for this argument. Indeed, I will assume that non-causal explanations are possible, which has been justified elsewhere within philosophical literature (Colyvan, 2001, 45-68). This is to allow that abstract mathematical objects can be used for explanation, as abstract objects cannot interact causally (Batterman, 2010, 2) (Baker, 2005, 228). Therefore, I take my conditions for mathematical explanations from Alan Baker, which I see as very plausible and intuitive, so will not defend them here (see Baker, 2005, 233-234 for justification and explanation of these conditions). The application of the mathematics in question must be external to mathematics, the phenomenon in question must be in need of explanation, the phenomenon must have been identified before the mathematics was used as explanation (so therefore it is more clearly explanation rather than prediction), and the mathematical component must be explanatory in its own right. I will therefore use this as a groundwork for the rest of the essay.

Mathematical Explanation in Science: A New Example

[3] For completeness, I will mention that there is ongoing debate within the scientific theory whether complex numbers may turn out to be necessary within our current formulation of quantum mechanics, alongside a proposal that this may be experimentally verifiable (Renou et al., 2021). By 'necessary', I do not wish to open this up to claims of dispensability but merely to note that complex numbers may be used within quantum mechanics in a different and more integral way than elsewhere. I will not discuss this further, however, as this claim has not yet been widely accepted within the scientific community.

I now wish to contribute a new example where mathematics is explanatory. In line with Baker's conditions, I will first outline the phenomena that was in need of explanation. Within the natural world, there are certain patterns that crop up frequently, from the number of petals on a flower (Minarova, 2014, 9, 14), to the arrangement of bracts in a pine cone (ibid., 10). Why, therefore, do certain numbers and patterns appear so frequently? For example, let us consider the following why-question, here adopting a 'why' model of explanation for fairly arbitrary reasons of clarity; why do asteraceas (daisies) mainly grow to have 34, 55, or 89 petals (ibid., 14)? To answer it with full explanatory power, it seems, requires reference to the Fibonacci series, and therefore mathematics is used in its explanation. The Fibonacci series is characterised by summation of the previous two numbers in the sequence, as follows:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89…

and so forth. Taking the ratio of a number in the sequence with its predecessor, as the sequence continues, the successive ratios tend towards a value of 1.618 (Gardiner, 2012, 1721). This is an irrational number, meaning it is a real number that cannot be expressed as the ratio of two integers, that is known as *phi*, or the 'golden ratio' (Livio, 2003, 4-6). Why is this set of integers important? It turns out that asteraceas have these numbers of petals because they are Fibonacci numbers, and these numbers represent one of the most efficient ways to develop and grow (Minarova, 2014, 9) – an appeal also to the theory of evolution, as well as to mathematics. This is the most effective growth pattern for the flowers, and thus the most efficient way to organise the petals to fill space. The evolution of plants therefore has resulted in asteraceas with numbers of petals that are Fibonacci numbers.

To defend against objections of question-begging, I will first note that the why-question as I have given it is phrased with reference to numbers, which some could argue already assumes abstract mathematical objects. However, this is for ease of understanding. I think that there are plausibly other ways in which this question could be posed without reference to numbers, roughly such as 'why do asteraceas follow certain shapes?', or some such thing, perhaps referring to space-time points rather than numbers. Indeed, I have only used a 'why' model for simplicity, and there are perhaps other (nominalised) ways to propose the explanandum. I believe that Baker has more fully and successfully developed a method for doing so for his own example of mathematical explanation (Baker, 2009, 619-622). Without wishing to defend this more completely in this essay, his method plausibly also extends to my own example.

To take this further, the mathematics is also explanatory (and perhaps more obviously so) if the why-question is expanded to include reference to other flowers. Why do buttercups and wild roses have 5 petals, delphiniums have 8 petals, and asters have 21 (Minarova, 2014, 14)? Again, I assume that there is a way to pose this question without making reference to abstract mathematical objects, and that they are not indispensable for asking such a question (Baker, 2009, 619-622), but this is the clearest and simplest form of such a question that I can imagine. Perhaps it would refer to patterns within nature. Why do so many flower species in nature have the same numbers of petals?[4] Part of the explanation of why certain patterns crop up so frequently in nature,[5] alongside mention 'this is the most effective or efficient way to arrange its constituent elements and conserve energy' or some such statement, and a statement on how species evolve, invokes mention of Fibonacci numbers. Thus, explaining the number of petals on asteraceas (or some other nominalised formulation of the question) requires mathematics.

To confirm the success of this example as an instance of mathematical explanation, I will run through Baker's conditions (Baker, 2009, 233-234). Firstly, the application of the mathematics within the explanation of the phenomenon must be external to mathematics. I think this quite obviously holds, as understanding why a flower has a certain number of petals seems quite external to mathematics. Secondly, the phenomenon in question must be in need of explanation. Again, I think this is clear; if somebody asked me why certain specific numbers of petals appear in various species, I would be quite curious to understand why. Thirdly, the phenomenon must have been identified before the explanation was found. Historically, this is true of my example, so there is no problem here (Livio, 2003, 109-113). It was certainly not predicted that flowers would commonly have numbers of petals that are Fibonacci numbers. Instead, the repeating patterns found in nature have been noted from as early as Theophrastus around 300 BC (Livio, 2003, 109-110), well before the discovery of Fibonacci numbers in the 13th Century (ibid., 92-98). As well as these, Baker finally asserts that the mathematics must be explanatory in its own right, and not just within a descriptive mathematical framework. This is based on a concern of Melia's that the mathematics that is indispensably used to explain a theory is still not adding to

[4] I would like here to acknowledge the work of Robert Batterman in discussing universality with regards to mathematical explanation, albeit in a different sense (see Batterman, 2010, 4-8).

[5] I use the example of numbers of petals rather than, say, shell patterns to avoid mention of geometry. I think it is more intuitive to point to explicit explanations that do not refer to geometry. This is because it seems more plausible that geometry could be nominalised, i.e. have any reference to abstract mathematical objects removed. For example, the case could be made that geometry refers to properties of physical objects (Baker, 2005, 228).

the theory's explanatory power (Melia, 2002, 76). However, as Baker discusses for his own example (Baker, 2005, 234-235), without developing a serious account of explanation, I think that the mathematics has explanatory power in answering a why-question, so it is very plausible that the mathematics itself is genuinely explanatory. Therefore, I think my example holds as an instance of mathematical explanation within biological science.

Initial Objections

I will now consider objections to this example. I believe that Baker has effectively countered many of the possible objections to his own example of a mathematical explanation in his 2009 work (Baker, 2009, 614-631), and that his arguments apply in defence of my example. For example, the charge of arbitrariness is, in my opinion, a fairly obvious objection. Essentially, this means that the mathematics used in the explanation could just as well be replaced by other objects, concepts or theories. This arbitrariness goes beyond choosing different representations for numbers such as saying 'pink + pink = purple' to replace '2+2=4'. I think that Baker, however, effectively demonstrates that at every level, choosing an example such as mine where the properties of numbers within the Fibonacci sequence play a key role within the explanation, and have a clear non-arbitrary physical basis, overcomes this charge of arbitrariness (ibid., 614-619). Due to space limitations, I will not flesh out how his arguments, and other arguments addressing further objections, directly apply to my example beyond this, as I think Baker has already done the important work.

Ultimately, could it be that Fibonacci numbers are not required to explain my why-questions? If we remove mention of numbers from the question, to avoid question-begging, then perhaps it is not very useful, and indeed indispensable, to refer to Fibonacci numbers in our explanation – one could just say that evolution causes the petals to arrange in the most efficient way.[6] I personally think that this causes the explanation to lose power, as it is less precise, and therefore the theory is worse without reference to abstract mathematical objects, despite perhaps requiring ontological commitment to fewer types of things. Going by my previous definition of indispensability, I therefore am of the opinion that reference of Fibonacci numbers is indispensable to the explanation. However, let us nonetheless proceed with the objection:

[6] Without having the space to develop this idea, I think that this objection as I develop it within the rest of the essay also applies to Baker's example.

invoking the theory of evolution, and that the petals evolve over time with the most efficient growth pattern, is sufficient (and more parsimonious) to explain the phenomenon and answer the why-question.

The Theory of Evolution

Let us therefore consider the theory of evolution more carefully, to see how it explains the phenomena. I think it is uncontroversial to note that the theory of evolution is one of our best scientific theories. When a member of a species has offspring, the genetic information passed from parent to child can undergo mutations, or simply result in the expression of different genetic information that seen in the parent (Mukherjee, 2016, 36-37). In a time or environment where certain traits are favourable for the survival of a species, the individuals of the species with favourable traits are more likely to survive. They are therefore more likely to pass on their favourable traits to their offspring (ibid., 37). Genetic expression (say, of favourable and unfavourable traits) is described by statistical distributions such as bell curves; applications of statistics within biology were significantly developed by the mathematical modelling work of Ronald Fisher (ibid., 103-104). These statistical approaches are used to explain how different traits are inherited to arise within a population. As well as this, the phenomenon of individuals of a species with favourable traits surviving in greater numbers than those without happens on average, so therefore can also be described statistically. There must be sufficient variation for evolution to occur (Mukherjee, 2016, 104). Here and throughout my essay, I use a simple definition of statistics as that which refers to statistical properties, such as averages, population distributions, and so forth[7] (Horan, 1994, 79). The importance of the probability and statistical distribution of a certain gene being expressed within a population, and the probability and statistical distribution of it being passed on to sufficient of its offspring to cause general evolution of the population, plausibly need to be understood to explain the theory of evolution.[8] A fairly intuitive indication of this is that the theory of evolution, generally speaking, is not concerned with individual organisms. There is a requirement that the theory

[7] I purposefully do not refer to mathematics here as the crucial question, as I will further discuss, will be whether statistical properties are necessarily properties relating to abstract mathematical objects!

[8] The statistical nature of the theory of evolution as I have described it is not necessarily the full picture. Various philosophers and scientists have attributed a statistical nature to the theory of evolution on different grounds, from natural selection and genetic expression as I have described to evolutionary dynamics (Horan, 1994, 76-78). I have presented what I think is a fairly non-controversial account of the theory of evolution, but I think it is important to note that I have not delved fully into all aspects of the theory (and, in my opinion, I could not do so to completeness within this space).

looks at and analyses populations, as requiring statistical analysis (Horan, 1994, 77). I think it is also important to note here that while understanding the theory of evolution on a surface level only requires a vague conception of statistical distributions and probability, more rigorous and scientific formulations presumably and plausibly require a more rigorous (and mathematical) conception.[9] I think that this is justified when considering what was required historically for the theory of evolution to be accepted within the scientific community (Mukherjee, 2016, 28-46, 56-63, 101-110), and the development of mathematical statistical laws for the theory (Wright, 1931, 202). As well as this, referring back to the meaning of indispensability as I previously discussed it, the theory of evolution can only dispense with statistics if a better theory could be developed without it. To conclude my point, to fully explain why-questions about the numbers of petals, if we refer to the theory of evolution and how the flowers evolved to grow in the most efficient manner, we also invoke reference to the statistical distributions of inheritance and of genetic expression. The question of how the flowers evolved is external to mathematics, needed to be explained, and was considered before application of statistical understandings within evolution.[10] Perhaps, therefore, explaining the theory of evolution refers to mathematical objects through statistics.

Statistics within the Indispensability Debate

This therefore leads us to consider the nature of statistics. The philosopher determined to assert that reference to abstract mathematical objects is dispensable within science has an obvious next move: to assert that statistics (notably, in the form that the theory of evolution requires to answer the why-question) can be nominalised. Again, I am slightly sceptical that statistics can be nominalised in this way to produce scientific theories that are better than those without reference to abstract mathematical objects. However, my intuitions are not sufficient to assert this rigorously, and will presumably be different to the intuitions of others. A philosopher like Hartry Field might be inclined to say that statistical properties are not relevant to the behaviour of the evolutionary system, so should not be used within our explanation as they are extrinsic rather than intrinsic features (Field, 1989, 193). It may even be sufficient to prove that statistics

[9] See Wright, 1931, 205-207 for an example of explicitly mathematical formulations within the theory of evolution.

[10] As I will state later in the essay, this example may not hold up so well to Melia's objection that statistics could just be a potential framework used for explanation, but I think debating this is part of the philosophical work that should be done further from this essay.

is just a descriptive framework used to explain the theory of evolution, in accordance with Melia's objection and Baker's conditions. I also think that statistics and probabilities could much more plausibly be removed from explanation within the theory of evolution than it can from, say, thermodynamics, which I will mention again later in the essay. Of course, the nominalised picture of statistics could also not make reference to abstract objects, such as via possible worlds which are generally considered abstract, or else it would be somewhat pointless. It is difficult to see how a theory that also refers to abstract objects would be better than one which uses abstract mathematical objects, or why, if other abstract objects could be given to exist, abstract mathematical objects could not. I do not consider myself capable of imagining a rigorous nominalised version of statistics, but this does not mean that it cannot be done. This could be a problem for a theory of evolution that only makes reference to abstract mathematical objects via statistics, and therefore to my example. Appealing to nominalised statistics could explain why a flower has a certain shape (i.e. number of petals), given that this shape is the most efficient way to grow petals, meaning the flower evolved to have the most efficient shape. If explanation from the theory of evolution is given to be sufficient for the numbers of petals on various flowers, and the theory of evolution does not indispensably make reference to abstract mathematical objects, then this is not an example of necessarily mathematical explanation that supports the indispensability argument.

On the other hand, if statistics necessarily utilises abstract mathematical objects, and explaining the theory of evolution requires statistics, my example holds. This is perhaps a little more difficult to prove, as I believe it might require ruling out other possible accounts of statistics, yet I do think it is compelling. If statistics could only be properly described using mathematics, and thus statistical explanations are mathematical explanations, then even the somewhat vague statistical conceptions I refer to within the theory of evolution utilise mathematical explanation. For my example, even if the explanation of the number of petals on an asteracea did not require reference to Fibonacci numbers, it would still require reference to the theory of evolution, which in turn indispensably requires some reference to statistics, which indispensably requires reference to abstract mathematical objects. In this case, if it is justified that the theory of evolution is required to explain why a flower has a certain number of petals, and we refer to the probabilities of certain properties and the statistical distribution of their expression in order to do so, it seems justified that we ought to believe in abstract mathematical objects, by virtue of the indispensability argument. In my opinion, then, it would be valuable to the indispensability

debate to further pursue the question of whether statistics indispensably uses abstract mathematical objects.

As well as this, even if one was able to argue that the theory of evolution does not indispensably require reference to statistics,[11] this could open up very compelling evidence for the indispensability argument within other areas of science. From quantum mechanics to modern thermodynamics, statistics appears frequently in scientific theory, and plausibly therefore in explanations for phenomena within science. The ramifications for what we ought to believe in when we use statistics within our best scientific theories – idealised entities and populations? abstract mathematical objects? possible worlds? – are intriguing. Therefore, I believe that further work should be done within the philosophy of mathematics to study the nature of statistics, and investigate the applications within other areas of science and scientific theories. Via the indispensability argument, this would offer further credible evidence that we ought to believe in abstract mathematical objects, if it could be argued that statistics within our best scientific theories necessarily refers to them.

Conclusion

In conclusion, affirming the explanatory nature of mathematics within science would be compelling evidence to accept the indispensability argument for the existence of abstract mathematical objects. I have provided my own example of where mathematics is explanatory to enrich the evidence for the indispensability argument. This was in answer to the question of why asteraceas have 34, 55 or 89 petals, by appeal to Fibonacci numbers and the theory of evolution. This can be extended to explain why both buttercups and wild roses have 5 petals per flower, and so forth. I have then argued that if one could determine that it is not necessary to use the Fibonacci sequence when answering such questions, even though I am sceptical of such a claim, the alternative explanation seems to refer to evolutionary theory, which uses statistics. Therefore, the status of statistics as necessarily mathematical (i.e. indispensably referring to abstract mathematical objects) should be determined. While I have not attempted to do so

[11] See Horan, 1994 for arguments that the theory of evolution is not a statistical theory. She also concludes that a better understanding of the statistical nature of the theory of evolution is needed, although from a different viewpoint than mine concerned with the philosophy of science. Interestingly, part of her argument is that using statistics is a problem for the explanatory power of the theory of evolution because statistics cannot form causal explanations (Horan, 1994, 81-94). Given that I accept that explanations can be non-causal, this would presumably not be a problem within my account; this could be an interesting line of thought to follow.

within this essay, I think that this is worthy of further philosophical efforts to make progress in the indispensability debate. This also could lead to further evidence of mathematical explanations in science via statistics, such as within thermodynamics.

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